Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		Subject: Stats for ML & DS
TE - 1 Batch: T13 Sem: 5	Roll No: 2401054	Date: 29/07/26

Experiment 2

Aim - To containerise applications using Docker

LO - Containerizer, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes and Cloud Infrastructure.

LO Mapped: LO4

Theory -

1. Introduction to Containerization

In conventional software deployment, developers frequently encounter the "it works on my machine" issue. This occurs because the development environment and the production environment may differ in terms of operating system versions, installed libraries, configuration settings, or dependency versions.

Containerization addresses this challenge by packaging an application together with all the libraries, dependencies, binaries, and configuration files it requires into a single isolated unit known as a **container**. As a result, the application behaves consistently and runs the same way across different computing environments.

2. What is Docker?

Docker is a containerization platform that packages an application along with its required libraries, dependencies, and configuration files into an isolated unit called a **container**. Unlike traditional virtual machines, Docker containers share the host operating system's kernel, making them lightweight and allowing them to start much faster. This ensures that applications execute consistently on any system, removing the common "it works on my laptop" deployment issue.

3. Docker Architecture

Docker follows a client-server architecture in which three major components work together:

[Docker Client] <----> [Docker Daemon (Host)] <----> [Docker Registry]

(CLI Commands) (Manages Containers) (Docker Hub)

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		**Subject: Stats for ML & DS**
TE - 1 Batch: T13 Sem: 5	**Roll No: 2401054**	**Date: 29/07/26**

1. **Docker Client (CLI):**
   The Docker Client acts as the main interface between the user and Docker. Whenever commands such as docker run, docker build, or docker rmi are executed from the terminal, the client forwards these instructions to the Docker Daemon for processing.

2. **Docker Host (Docker Daemon / dockerd):**
   The Docker Daemon is a continuously running background service that receives requests through the Docker API. It is responsible for tasks such as creating images, managing containers, setting up networking, and handling storage volumes.

3. **Docker Registry:**
   A Docker Registry serves as a centralized repository for storing Docker images. Docker Hub is the default public registry where developers can publish and download ready-made images for operating systems, databases, programming languages, and various software tools.

4. Core Components of Docker

To understand Docker properly, it is important to know its three primary components:

- **Docker Image:**
  A Docker image is a read-only and immutable template generated from a Dockerfile. It includes the application's source code, required libraries, dependencies, and runtime environment. Images are built using layered architecture, which helps optimize storage usage. An image itself cannot execute; instead, it acts as the foundation for creating containers.

- **Docker Container:**
  A Docker container is a running instance of a Docker image. It introduces a thin writable layer over the image's read-only layers, enabling the application to create files, store data, and execute processes while remaining isolated from the host system and other containers.

Output:

1) What does docker images do?

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		**Subject: Stats for ML & DS**
TE - 1 Batch: T13 Sem: 5	**Roll No: 2401054**	**Date: 29/07/26**

Ans: It serves as a "Ready-to-Run" Library: Because these images are listed, they are already downloaded or built on your machine. You can launch a container from any of them immediately without needing an internet connection to pull them from a registry.

```
PS C:\Users\15L> docker images

IMAGE                     ID             DISK USAGE   CONTENT SIZE   EXTRA
httpd:latest              4613a77dcb46       117MB             0B   U
mysql:latest              a88c3e85e887       632MB             0B
new_image_name:latest     df8b6090dd41       140MB             0B
nginx:1.0                 fcbd66995946       169MB             0B   U
node:latest               c3978d05bc68       1.1GB             0B   U
redis:latest              3848bb827e71       140MB             0B   U
sample-flask-app:v1       657cf5304039       141MB             0B   U
ubuntu:latest             de7345b16e94       100MB             0B   U
```

2) What does docker rmi new_image_name:latest does?

Ans: The docker rmi command is used to delete a specified Docker image from the local system, helping recover storage space. In this case, it removed the image tagged as new_image_name:latest by untagging it and deleting its associated image layers. The extra . at the end of the command was not required, since the docker rmi command only needs the image name or image ID to remove the image.

```
PS C:\Users\15L> docker rmi new_image_name:latest .
Untagged: new_image_name:latest
Deleted: sha256:df8b6090dd41b04c834722a31f288ca31c82d129c4843f8912b6dfa3086ec379
Deleted: sha256:fdb84ffd6f82527ed418183e8d2f68a59ed3ef74a01760a3c5c0ccb56490341c
```

3) What does docker image prune do?

Ans: The docker image prune command is used to clean up the local Docker environment by removing **dangling images**, thereby freeing up disk space.

A **dangling image** is an image that no longer has a valid tag (for example, it does not have a name such as my-app:latest) and is not associated with any existing container. Such images are commonly created when a new version of an image is built using the same tag, leaving the previous version untagged and unused on the system.

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		**Subject: Stats for ML & DS**
TE - 1 Batch: T13 Sem: 5	**Roll No: 2401054**	**Date: 29/07/26**

```
PS C:\Users\15L> docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] y
Deleted Images:
untagged: redis@sha256:3134997edb04277814aa51a4175a588d45eb4299272f8eff2307bbf8b39e4d43
deleted: sha256:170a1e90f8436daa6778aeea3926e716928826c215ca23a8dfd8055f663f9428
deleted: sha256:07a225c671a5a8f3c0d3f48590dbb3c9567d00a020c49fff9e60d8dbd7fdc925
deleted: sha256:ea580ef52fceb9e8458910388ea2b0cb7793a252bd5ece4a6fbdaee97b052ccd
deleted: sha256:36022d92671a31dbf2801a15735a4cddfe220f755e6dfc1f094faacf8d4d68ec
deleted: sha256:5ab673946d6214ee4cca972472c8504cb7e2f606cde16d7bdc81b995f33532cc
deleted: sha256:b4b7f9d393a71ddd2d4ee4d08c9b0abb5494158126545b5907d559b1530d8e1b8
deleted: sha256:a73fbd0cf0721f58760274cdf40771a5380fdacd993e5448b5dfab978e06e745
deleted: sha256:da36ef59ebafc4af9e78fce5d6ff603f937befdb2270d3733dd25aa39cb8c563
deleted: sha256:a483da8ab3e941547542718cacd3258c6c705a63e94183c837c9bc44eb608999
untagged: ubuntu@sha256:f9d633ff6640178c2d0525017174a688e2c1aef28f0a0130b26bd5554491f0da
deleted: sha256:3db8720ecbf5f5927d409cc61f9b4f7ffe23283917caaa992f847c4d83338cc1
deleted: sha256:d101c9453715a978a2a520f553588e77dfb4236762175eba61c5c264a449c75d
```

DOCKER HUB:

4) How can I log into Docker Hub from my terminal using a personal access token?

Ans: By using the echo "TOKEN" | docker login -u username --password-stdin format, you are securely passing your password/token into the command without typing it directly into the terminal history, where it might be seen by others.

```
PS C:\Users\15L> echo "dckr_pat_QxVxZE4T9vUEnH5_ZcR-mbofdD8" | docker login -u ayush1248 --password-stdin
Login Succeeded
```

5) What does docker mysql(image) does?

Ans: The docker search command queries the Docker Hub registry to find images matching your specified keyword. The output provides a ranked list displaying the repository name, a brief description, community popularity through "stars," and whether it is an official, verified image. This tool helps you evaluate and choose the most reliable and trusted images before downloading them to your local environment.

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		Subject: Stats for ML & DS
TE - 1 Batch: T13 Sem: 5	Roll No: 2401054	Date: 29/07/26

```
PS C:\Users\15L> docker search mysq
NAME                              DESCRIPTION                                     STARS     OFFICIAL
mysql                             MySQL is a widely used, open-source relation…   16157     [OK]
utri1010/mysq                                                                     0
rumpl/mysq                                                                        0
micheken20/mysq                                                                   0
praveenkumar2608/mysq                                                             0
yad87/mysq                                                                        0
tyft/mysq                                                                         0
raaghavendra/mysq                                                                 0
circleci/mysql                    MySQL is a widely used, open-source relation…   33
ryo113/mysq                                                                       0
cimg/mysql                                                                        4
losged/mysq                                                                       0
22211tt4921vinhnguyen/mysq                                                        0
esha0629/mysq                                                                     0
naveen2809/mysq                   dpt mysql                                       0
21yjs/mysq                        我的私人仓库                                    0
bitnamicharts/mysql               Bitnami Helm chart for MySQL                    1
zabbix/zabbix-web-nginx-mysql     Zabbix frontend based on Nginx web-server wi…   267
zabbix/zabbix-server-mysql        Zabbix Server with MySQL database support       432
zabbix/zabbix-proxy-mysql         Zabbix proxy with MySQL database support        48
airbyte/source-mysql                                                              0
ubuntu/mysql                      MySQL open source fast, stable, multi-thread…   74
zabbix/zabbix-web-apache-mysql    Zabbix frontend based on Apache web-server w…   78
zabbix/zabbix-build-mysql         Zabbix build base (MySQL)                       6
google/mysql                      MySQL server for Google Compute Engine          26
```

6) What does docker pull mysql(image)?

Ans: The docker pull mysql command downloads the official MySQL database image from the Docker Hub registry to your local machine. Once the download is complete, the image is stored in your local repository, making it ready to be used to create and run database containers. This process ensures you have the necessary blueprint saved locally to launch instances of MySQL without needing to re-download the image each time.

```
PS C:\Users\15L> docker pull mysql
Using default tag: latest
latest: Pulling from library/mysql
c2fb254fc2f4: Pull complete
e21df1635f12: Pull complete
88745f4755f0: Pull complete
08b808cb6d40: Pull complete
e0607e526570: Pull complete
61d49498d3da: Pull complete
5ac73e0e3f6a: Pull complete
bb49c3208fe7: Pull complete
1576763b528a: Pull complete
81b3a6a3bce8: Pull complete
Digest: sha256:8fdf311514c91fa5014e93e98e19d7f5d9eb162a462c13100c956dacc278ee21
Status: Downloaded newer image for mysql:latest
docker.io/library/mysql:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview mysql
```

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		Subject: Stats for ML & DS
TE - 1 Batch: T13 Sem: 5	Roll No: 2401054	Date: 29/07/26

```
PS C:\Users\15L> docker tag sample-flask-app:v1 ayush1248/sample-flask-app:v1
TS C:\Users\15L> docker push ayush1248/sample-flask-app:v1
The push refers to repository [docker.io/ayush1248/sample-flask-app]
fa81a3d49760: Layer already exists
f60370cada3c: Layer already exists
785aa2d00513: Layer already exists
8347a0657a00: Layer already exists
34b0037810a0: Layer already exists
188695d9eb1d: Layer already exists
188c9b34dfbe: Layer already exists
v1: digest: sha256:ab3c9f34dfbefa81a3d49760f60370cada3c785aa2d005138347a0657a0034b003781
size: 1596
```

GENERAL COMMANDS:

```
PS C:\Users\15L> docker run --help
Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Create and run a new container from an image

Aliases:
  docker container run, docker run

Options:
      --add-host list                  Add a custom host-to-IP mapping
                                       (host:ip)
      --annotation map                 Add an annotation to the
                                       container (passed through to the
                                       OCI runtime) (default map[])
  -a, --attach list                    Attach to STDIN, STDOUT or STDERR
      --blkio-weight uint16            Block IO (relative weight),
                                       between 10 and 1000, or 0 to
                                       disable (default 0)
      --blkio-weight-device list       Block IO weight (relative device
                                       weight) (default [])
      --cap-add list                   Add Linux capabilities
      --cap-drop list                  Drop Linux capabilities
      --cgroup-parent string           Optional parent cgroup for the
                                       container
      --cgroupns string                Cgroup namespace to use
                                       (host|private)
                                       'host':    Run the container in
                                       the Docker host's cgroup
                                       namespace
                                       'private': Run the container in
                                       its own private cgroup namespace
                                       '':        Use the cgroup
                                       namespace as configured by the
                                                  default-cgroupns-mode
                                       option on the daemon (default)
      --cidfile string                 Write the container ID to the file
      --cpu-period int                 Limit CPU CFS (Completely Fair
                                       Scheduler) period
```

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		Subject: Stats for ML & DS
TE - 1 Batch: T13 Sem: 5	Roll No: 2401054	Date: 29/07/26

```
                                        (0-3, 0,1)
  -d, --detach                          Run container in background and
                                        print container ID
      --detach-keys string              Override the key sequence for
                                        detaching a container
      --device list                     Add a host device to the container
      --device-cgroup-rule list         Add a rule to the cgroup allowed
                                        devices list
      --device-read-bps list            Limit read rate (bytes per
                                        second) from a device (default [])
      --device-read-iops list           Limit read rate (IO per second)
                                        from a device (default [])
      --device-write-bps list           Limit write rate (bytes per
                                        second) to a device (default [])
      --device-write-iops list          Limit write rate (IO per second)
                                        to a device (default [])
      --dns list                        Set custom DNS servers
      --dns-option list                 Set DNS options
      --dns-search list                 Set custom DNS search domains
      --domainname string               Container NIS domain name
      --entrypoint string               Overwrite the default ENTRYPOINT
                                        of the image
  -e, --env list                        Set environment variables
      --env-file list                   Read in a file of environment
                                        variables
      --expose list                     Expose a port or a range of ports
      --gpus gpu-request                GPU devices to add to the
                                        container ('all' to pass all GPUs)
      --group-add list                  Add additional groups to join
      --health-cmd string               Command to run to check health
      --health-interval duration        Time between running the check
                                        (ms|s|m|h) (default 0s)
      --health-retries int              Consecutive failures needed to
                                        report unhealthy
      --health-start-interval duration  Time between running the check
                                        during the start period
                                        (ms|s|m|h) (default 0s)
      --health-start-period duration    Start period for the container
                                        to initialize before starting
                                        health-retries countdown
                                        (ms|s|m|h) (default 0s)
      --health-timeout duration         Maximum time to allow one check
                                        to run (ms|s|m|h) (default 0s)
      --help                            Print usage
  -h, --hostname string                 Container host name
      --init                            Run an init inside the container
```

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		Subject: Stats for ML & DS
TE - 1 Batch: T13 Sem: 5	Roll No: 2401054	Date: 29/07/26

```
-i, --interactive                  Keep STDIN open even if not attached
    --ip ip                        IPv4 address (e.g., 172.30.100.104)
    --ip6 ip                       IPv6 address (e.g., 2001:db8::33)
    --ipc string                   IPC mode to use
    --isolation string             Container isolation technology
-l, --label list                   Set meta data on a container
    --label-file list              Read in a line delimited file of
                                   labels
    --link list                    Add link to another container
    --link-local-ip list           Container IPv4/IPv6 link-local
                                   addresses
    --log-driver string            Logging driver for the container
    --log-opt list                 Log driver options
    --mac-address string           Container MAC address (e.g.,
                                   92:d0:c6:0a:29:33)
-m, --memory bytes                 Memory limit
    --memory-reservation bytes     Memory soft limit
    --memory-swap bytes            Swap limit equal to memory plus
                                   swap: '-1' to enable unlimited swap
    --memory-swappiness int        Tune container memory swappiness
                                   (0 to 100) (default -1)
    --mount mount                  Attach a filesystem mount to the
                                   container
    --name string                  Assign a name to the container
    --network network              Connect a container to a network
    --network-alias list           Add network-scoped alias for the
                                   container
    --no-healthcheck               Disable any container-specified
                                   HEALTHCHECK
    --oom-kill-disable             Disable OOM Killer
    --oom-score-adj int            Tune host's OOM preferences
                                   (-1000 to 1000)
    --pid string                   PID namespace to use
    --pids-limit int               Tune container pids limit (set
                                   -1 for unlimited)
    --platform string              Set platform if server is
                                   multi-platform capable
    --privileged                   Give extended privileges to this
                                   container
-p, --publish list                 Publish a container's port(s) to
                                   the host
-P, --publish-all                  Publish all exposed ports to
                                   random ports
```

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		Subject: Stats for ML & DS
TE - 1 Batch: T13 Sem: 5	Roll No: 2401054	Date: 29/07/26

```
                                        container
  -p, --publish list                    Publish a container's port(s) to
                                        the host
  -P, --publish-all                     Publish all exposed ports to
                                        random ports
      --pull string                     Pull image before running
                                        ("always", "missing", "never")
                                        (default "missing")
  -q, --quiet                           Suppress the pull output
      --read-only                       Mount the container's root
                                        filesystem as read only
      --restart string                  Restart policy to apply when a
                                        container exits (default "no")
      --rm                              Automatically remove the
                                        container and its associated
                                        anonymous volumes when it exits
      --runtime string                  Runtime to use for this container
      --security-opt list               Security Options
      --shm-size bytes                  Size of /dev/shm
      --sig-proxy                       Proxy received signals to the
                                        process (default true)
      --stop-signal string              Signal to stop the container
      --stop-timeout int                Timeout (in seconds) to stop a
                                        container
      --storage-opt list                Storage driver options for the
                                        container
      --sysctl map                      Sysctl options (default map[])
      --tmpfs list                      Mount a tmpfs directory
  -t, --tty                             Allocate a pseudo-TTY
      --ulimit ulimit                   Ulimit options (default [])
      --use-api-socket                  Bind mount Docker API socket and
                                        required auth
  -u, --user string                     Username or UID (format:
                                        <name|uid>[:<group|gid>])
      --userns string                   User namespace to use
      --uts string                      UTS namespace to use
  -v, --volume list                     Bind mount a volume
      --volume-driver string            Optional volume driver for the
                                        container
      --volumes-from list               Mount volumes from the specified
                                        container(s)
  -w, --workdir string                  Working directory inside the
                                        container
```

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		Subject: Stats for ML & DS
TE - 1 Batch: T13 Sem: 5	Roll No: 2401054	Date: 29/07/26

```
PS C:\Users\15L> docker info
Client:
 Version:    29.6.1
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.98.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.27.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.35.0-desktop.2
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.3.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.4.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.5
    Path:     C:\Program Files\Docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.43.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-mcp.exe
  model: Docker Model Runner (Docker Inc.)
    Version:  v1.2.5
    Path:     C:\Program Files\Docker\cli-plugins\docker-model.exe
  offload: Docker Offload (Docker Inc.)
    Version:  v0.6.7
    Path:     C:\Program Files\Docker\cli-plugins\docker-offload.exe
  pass: Docker Pass Secrets Manager Plugin (beta) (Docker Inc.)
    Version:  v0.2.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-pass.exe
  sandbox: "docker sandbox" is deprecated, use Docker Sandboxes instead (Docker Inc.)
    Version:  v0.13.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-sandbox.exe
  scout: Docker Scout (Docker Inc.)
```

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		Subject: Stats for ML & DS
TE - 1 Batch: T13 Sem: 5	Roll No: 2401054	Date: 29/07/26

```
Server:
 Containers: 30
  Running: 2
  Paused: 0
  Stopped: 28
 Images: 11
 Server Version: 29.6.1
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi
  /var/run/cdi
 Discovered Devices:
  cdi: docker.com/gpu=webgpu
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 nvidia runc
 Default Runtime: runc
 Init Binary: docker-init
 containerd version: e53c7c1516c3b2bff98eb76f1f4117477e6f4e66
 runc version: v1.3.6-0-g491b69ba
 init version: de40ad0
 Security Options:
  seccomp
   Profile: builtin
  cgroupns
 Kernel Version: 6.18.33.2-microsoft-standard-WSL2
 Operating System: Docker Desktop
 OSType: linux
 Architecture: x86_64
 CPUs: 20
 Total Memory: 15.49GiB
 Name: docker-desktop
 ID: 8426088b-7a55-4da0-a9e7-301b5465a3fd
 Docker Root Dir: /var/lib/docker
 Debug Mode: false
 HTTP Proxy: http.docker.internal:3128
 HTTPS Proxy: http.docker.internal:3128
 No Proxy: hubproxy.docker.internal
 Labels:
  com.docker.desktop.address=npipe://\\.\pipe\docker_cli
```

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		**Subject: Stats for ML & DS**
TE - 1 Batch: T13 Sem: 5	**Roll No: 2401054**	**Date: 29/07/26**

```
No Proxy: hubproxy.docker.internal
Labels:
 com.docker.desktop.address=npipe://\\.\pipe\docker_cli
Experimental: false
Insecure Registries:
 hubproxy.docker.internal:5555
 ::1/128
 127.0.0.0/8
Live Restore Enabled: false
Firewall Backend: iptables
```

CONTAINERS:



"How do I create, manage, and interact with a Docker container using the command line?"

Ans: **docker run -d redis**: Creates and starts a new container from the 'redis' image in detached mode (-d), running it in the background and returning a container ID.

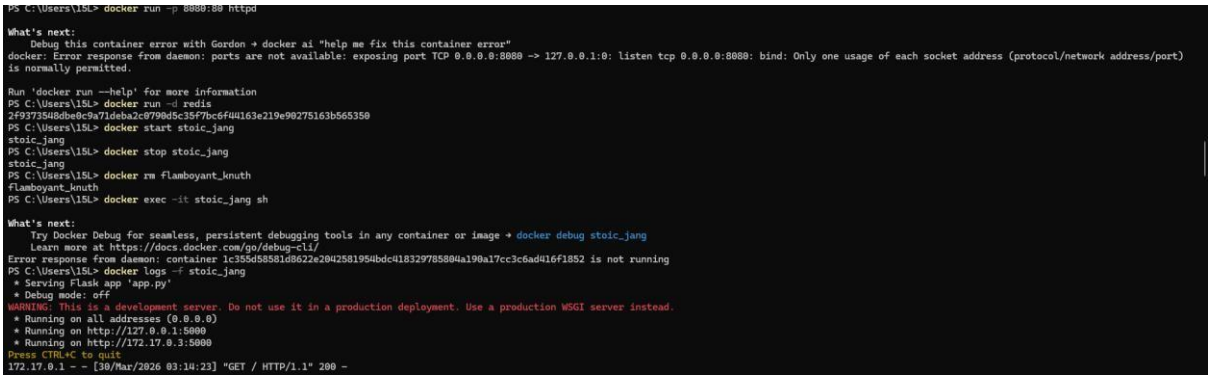
docker start / docker stop / docker rm: These commands manage the container's lifecycle by starting a stopped container, stopping a running one, or removing an existing container from the system.

docker exec -it stoic_jang sh: Opens an interactive (-it) shell (sh) inside the running container named 'stoic_jang', allowing you to execute commands directly within that container's environment.

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		**Subject: Stats for ML & DS**
TE - 1 Batch: T13 Sem: 5	**Roll No: 2401054**	**Date: 29/07/26**



How can I view the full technical configuration, network settings, and current status of a specific Docker container?

Ans: Detailed Information: The command displays a complete JSON output containing information such as the container ID, creation time, executable path, environment variables, and startup parameters.

Container Status: It provides the current state of the container (for example, "Status": "exited"), indicating whether the container is running or stopped, along with details such as the last ExitCode (e.g., 137).

Configuration Details: It also includes important system information such as network settings, logging locations, mounted volumes, and platform-specific configuration, which are useful for troubleshooting and understanding how the container was configured.

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha		**Subject: Stats for ML & DS**
TE - 1 Batch: T13 Sem: 5	**Roll No: 2401054**	**Date: 29/07/26**

```
PS C:\Users\15L> docker inspect stoic_jang
[
    {
        "Id": "1c355d58581d8622e2042581954bdc41832978580 4a190a17cc3c6ad416f1852",
        "Created": "2026-03-30T03:14:02.301738642Z",
        "Path": "flask",
        "Args": [
            "run",
            "--host=0.0.0.0",
            "--port=5000"
        ],
        "State": {
            "Status": "exited",
            "Running": false,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 0,
            "ExitCode": 137,
            "Error": "",
            "StartedAt": "2026-07-20T05:20:52.348899794Z",
            "FinishedAt": "2026-07-20T05:21:02.907806571Z"
        },
        "Image": "sha256:657cf5304039e00f787e6866a6771740c1aa630272e9c5d748cbd5e67eb5ddb4",
        "ResolvConfPath": "/var/lib/docker/containers/1c355d58581d8622e2042581954bdc41832978580 4a190a17cc3c6ad416f1852/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/1c355d58581d8622e2042581954bdc41832978580 4a190a17cc3c6ad416f1852/hostname",
        "HostsPath": "/var/lib/docker/containers/1c355d58581d8622e2042581954bdc41832978580 4a190a17cc3c6ad416f1852/hosts",
        "LogPath": "/var/lib/docker/containers/1c355d58581d8622e2042581954bdc41832978580 4a190a17cc3c6ad416f1852/1c355d58581d8622e2042581954bdc41832978580 4a190a17cc3c6ad416f1852-json.log",
        "Name": "/stoic_jang",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "bridge",
            "PortBindings": {
```

How can I view a list of all containers that are actively running on my system?

Ans:

- **Active Monitoring**: The command displays key information for every container that is currently in an "Up" status.
- **Essential Details**: It shows vital identifiers like the **CONTAINER ID** and the **NAME** assigned to each container, along with the **IMAGE** it was built from.
- **Operational Context**: It reveals how long each container has been running, the command used to start it, and any network **PORTS** currently exposed to the outside.

```
PS C:\Users\15L> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS      NAMES
2f9373548dbe   redis     "docker-entrypoint.s…"   5 minutes ago    Up 5 minutes    6379/tcp   sad_cartwright
c765c3d83e84   ubuntu    "/bin/bash"              51 minutes ago   Up 51 minutes              hungry_blackburn
b371b47ddd31   ubuntu    "/bin/bash"              54 minutes ago   Up 54 minutes              distracted_clarke
```

Thadomal Shahani Engineering College, Bandra

Department of Artificial Intelligence & Data Science

Name: Ayush Kushwaha			**Subject: Stats for ML & DS**
TE - 1 Sem: 5	**Batch: T13**	**Roll No: 2401054**	**Date: 29/07/26**

How can I view a complete list of all my containers, including those that have exited or are just in the 'Created' state?

Ans:

- **Comprehensive Inventory**: Unlike `docker ps`, which only shows active containers, the `--all` flag (or `-a`) displays every container record currently stored on your machine.
- **Status Visibility**: It allows you to see the current state of your environment, such as which containers have `Exited` (with specific exit codes like 0 or 137) and which are merely `Created` but never started.
- **History & Cleanup**: This view is essential for auditing your development history or identifying old, unused containers that are consuming system resources and should be pruned.

```
PS C:\Users\15L> docker ps --all
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS                       PORTS                  NAMES
2f9373548dbe   redis                "docker-entrypoint.s…"   6 minutes ago    Up 6 minutes                 6379/tcp               sad_cartwright
885d5e85d231   httpd                "httpd-foreground"       6 minutes ago    Created                                             reverent_snyder
452bcec1feb4   sample-flask-app:v1  "flask run --host=0.…"   6 minutes ago    Exited (0) 6 minutes ago                            my_new_app
c765c3d83e84   ubuntu               "/bin/bash"              52 minutes ago   Up 52 minutes                                       hungry_blackburn
b371b47ddd31   ubuntu               "/bin/bash"              55 minutes ago   Up 55 minutes                                       distracted_clarke
5ba8b97fc7a2   ubuntu               "\\bin\\bash"            56 minutes ago   Created                                             focused_wilson
1c355d58581d   sample-flask-app:v1  "flask run --host=0.…"   3 months ago     Exited (137) 5 minutes ago                          stoic_jang
498c2de19d41   redis                "docker-entrypoint.s…"   3 months ago     Exited (255) About an hour ago   6379/tcp           vibrant_goldberg
e81c8ac65cd6   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             jolly_herschel
f90bf8584ef0   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             inspiring_montalcini
c1558a5bdfc7   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             compassionate_cerf
61c302b4db0a   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             suspicious_heisenberg
a916b1bc1d13   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             nice_roentgen
004bcde700ff   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             sad_noyce
c8cb01650812   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             flamboyant_pike
d60547f1917e   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             intelligent_greider
05e923e0452c   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             zealous_bouman
250bb0ef6efe   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             sad_elion
74b68ec74672   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             hungry_jang
55a80be3d445   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             youthful_matsumoto
4ed8cfaded6b   httpd                "httpd-foreground"       8 months ago     Exited (0) 8 months ago                             sleepy_gates
2fa633478597   ca2b0f26964c         "/bin/bash"              2 years ago      Exited (255) 10 months ago                          unruffled_wilson
ee37479b71cd   node                 "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                              sarth
9c7ca042694c   node                 "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                              lucid_faraday
f4b59b6cf1f4   d1397258b209         "docker-entrypoint.s…"   2 years ago      Exited (255) 10 months ago       6379/tcp           musing_archimedes
d820ef6a1a06   d2c94e258dcb         "/hello"                 2 years ago      Exited (0) 2 years ago                              bb71b1a51783
bb71b1a51783   d2c94e258dcb         "/hello"                 2 years ago      Exited (0) 2 years ago                              silly_wozniak
922a7dec7144   d1397258b209         "docker-entrypoint.s…"   2 years ago      Exited (255) 2 years ago         6379/tcp           festive_hertz
97abe06c14c7   d2c94e258dcb         "/hello"                 2 years ago      Exited (0) 2 years ago                              b9f4789574a5
ae743b95da80   d2c94e258dcb         "/hello"                 2 years ago      Exited (0) 2 years ago                              mystifying_ramanujan
b9f4789574a5   d2c94e258dcb         "/hello"                 2 years ago      Exited (0) 2 years ago                              interesting_yalow
a21892131937   nginx:1.0            "/usr/sbin/nginx -g …"   2 years ago      Exited (255) 2 years ago         0.0.0.0:9090->80/tcp   webserver
PS C:\Users\15L>
```

Conclusion:

Docker offers an efficient client-server architecture that resolves common deployment issues such as the "it works on my machine" problem. It enables developers to package applications together with all necessary dependencies into lightweight, isolated, and portable containers,

Name: Ayush Kushwaha		**Subject: Stats for ML & DS**
TE - 1 Batch: T13 Sem: 5	**Roll No: 2401054**	**Date: 29/07/26**

ensuring consistent execution across different environments while providing a powerful collection of command-line tools for container management.